

When AI Misleads: Human Decision Failures in Hiring

Section I: Introduction

Artificial intelligence (AI) now plays a central role in hiring and recruitment, influencing decisions once made solely by human evaluators. Organizations increasingly rely on AI to screen resumes, rank candidates, analyze interviews, and predict job performance—promising greater efficiency, objectivity, and fairness. Yet research shows that AI hiring systems do not correct bias but instead mirror biases embedded in their training data, and these outcomes are reinforced when users rely on the system's recommendations without careful scrutiny (Wilson et al., 2025; Rosenthal-von der Pütten & Sach, 2024). Instead of removing bias, AI can conceal it behind a facade of neutrality, making discriminatory outcomes harder to detect or challenge.

The hiring process involves multiple decision-makers whose judgments are shaped by AI in different ways. HR managers remain the formal gatekeepers of candidate selection, adopting AI tools to increase efficiency, reduce subjectivity, ensure fairness, and identify the most qualified candidates. Their choices, however, are increasingly influenced—sometimes constrained—by algorithmic recommendations. Developers and executives shape outcomes as well by determining how models are trained, what data they learn from, and how recommendations are integrated into workflow. However, AI tools often fail to fully achieve the efficiency, fairness, and objectivity that evaluators expect, instead weakening human oversight by shaping judgments in ways they do not fully recognize. This occurs not only because of flawed algorithms but also because human decision makers remain vulnerable to well-documented cognitive biases—automation bias, overconfidence, anchoring, and the illusion of control—that distort how they interpret AI guidance (Kupfer et al., 2023; Soleimani et al., 2022).

AI in hiring processes is becoming an increasing problem because these systems now significantly influence who is screened, shortlisted, and ultimately considered for job

opportunities. By embedding structural inequities from their training data, AI tools can amplify discrimination at scale (Chen, 2023). When humans perceive AI outputs as objective, the system's influence becomes even harder to scrutinize, fostering a dangerous sense of “illusory fairness” (Kupfer et al., 2023). As humans defer to algorithmic outputs, accountability blurs—raising the question of whether responsibility lies with the system or its human users (Rosenthal-von der Pütten & Sach, 2024).

Addressing these challenges requires more than technical fixes. Biases emerge from the complex interplay between algorithms and human cognition (Soleimani et al., 2022). To ensure fairer hiring practices, decision makers must understand how AI shapes human decision-making and apply insights from decision science to design systems that promote transparency, oversight, and genuine equity.

Section II: Current approaches

Nowadays, AI-driven hiring has been promoted as a way to streamline screening and reduce subjectivity, yet current systems often amplify existing biases by evaluating superficial indicators—such as keywords, linguistic patterns, or appearance—rather than holistic qualifications. This effect creates an environment where even slight model preferences disproportionately influence who receives interviews and is ultimately hired. Therefore, AI in hiring can be harmful and significantly affect individuals' career opportunities.

Evidence of these biases is well documented. A University of Washington study found that LLMs favored white-associated names 85% of the time, female-associated names only 11% of the time, and never favored Black male names over white male names (Milne, 2024). These disparities illustrate how biased LLM outputs can translate into unequal hiring outcomes when decision makers do not scrutinize them carefully, ultimately leading to unfair outcomes based on

gender, race, and other identity makers (Chen, 2024). Because these problems carry serious consequences for applicants, several approaches have emerged to address bias in AI-assisted hiring. Current solutions generally fall into three categories: legal and policy, organizational, and technical. Yet each falls short in important ways.

Legal and policy approaches across the US focus on transparency and accountability. New York City's Automated Employment Decision Tool (AEDT) law requires that employers conduct annual bias audits and disclose when AI is being used to applicants. Similarly, Illinois' AI Video Interview Act mandates both disclosures and data retention guidelines (Kempe, 2024). These laws demonstrate that governments can regulate AI bias and promote greater accountability, and the EEOC has clarified that longstanding discrimination laws (Title VII, ADA, ADEA) apply to automated systems, meaning employers are legally required to test, monitor, and correct discriminatory AI outcomes (Kempe, 2024). However, these policies remain limited. Most regulations rely on self-reported audits, which can vary greatly, and without meaningful enforcement mechanisms, employers face little pressure to report problems accurately. This challenge is compounded by psychological tendencies: when decision makers overestimate their ability to detect bias or assume the system is already objective, then they may not view audits as necessary or take them seriously. Simply stating that AI is used does not alter the underlying decision-making process or address how easily evaluators overtrust automated tools. As a result, legal and policy interventions increase transparency but still fall short of ensuring fairness.

At the organizational level, employers have focused on adopting internal policies intended to balance automation with human oversight. Many companies state that recruiters "validate" AI recommendations or review each applicant's HireVues before making advancement decisions. This hybrid approach is shown through a talent-matching platform called Index.dev, which

connects employers to engineering talent. In their hiring model, they state that AI tools filter and score applicants, but “human involvement is essential to ensure fairness” through follow-up questions and soft-skill evaluations (Premkumar, 2025). Similarly, HR companies such as HireVue and Pymetrics report collaborating with organizational psychologists to develop and validate assessments (Premkumar, 2025). These collaborations aim to ensure that analysis aligns with established ethical standards. However, these strategies do not meaningfully change how recruiters interact with AI. Because reviewers typically see the AI’s scores or rankings first, their judgments are shaped around this initial reference point. Internal policies can also create an illusion of control: organizations feel reassured by expert involvement or ethics committees, yet recruiters still defer to automated recommendations in practice. As a result, organizational policies signal oversight but leave the underlying cognitive vulnerabilities largely unaddressed.

At the technical level, efforts to reduce AI bias in hiring primarily target training data and model evaluation. Many screening systems are built on historical datasets (i.e. resumes of past or current employees) which means the algorithm learns and reproduces existing demographic patterns, such as predominantly white and male workforces. Because these models are trained on imbalanced data, they replicate the same subjective patterns recruiters aim to eliminate. Developers sometimes attempt to correct this through data-balancing strategies. For example, Microsoft improved its Face API by revising its dataset, achieving a “20-fold reduction in the recognition error ratio between men and women with darker skin tones and a 9-fold reduction for women by balancing factors such as skin color, age, and gender” (“AI in hiring”). But these technical solutions address symptoms more than the root cause of bias. Removing demographic indicators does not on its own eliminate bias because models still infer these traits through features like names and vocabulary. Attempts to improve datasets can also inadvertently heighten

automation bias, as decision makers become more confident in AI outputs and less likely to scrutinize them. Despite growing awareness of these issues, current approaches remain limited because they fail to address the cognitive biases that shape how decision makers interpret AI output, leaving the core fairness problems unresolved.

Section III: Applying decision science

What is most concerning about the current approaches to AI-assisted hiring is that people often fail to recognize the extent to which AI systems introduce harm and reinforce bias, and even more critically, how their own cognitive tendencies cause them to trust and rely on these systems far more than they should. The first and perhaps most visible mechanism is automation bias. Automation bias is the tendency of humans to over-rely on automated systems' recommendations, leading to uncritical acceptance of AI-generated advice, even when that advice may be flawed or biased (Parasuraman & Manzey, 2010, p. 391). In hiring, this occurs when evaluators treat AI-generated scores as objective indicators of merit. Behavioral evidence clearly illustrates this dynamic. Wilson et al. (2025) found that participants altered their hiring decisions to align with biased AI rankings up to 90% of the time, despite making more equitable selections when evaluating candidates independently. This reflects a classic automation bias pattern: the presence of an automated recommendation suppresses careful information processing, leading individuals to defer to AI decisions even when those decisions amplify discrimination. The study shows that automation does not simply support human judgment—it can override it entirely.

A second mechanism that reinforces this reliance on AI is anchoring, the cognitive bias in which people's judgments gravitate toward an initial piece of information and remain insufficiently adjusted from that starting point (Kahneman, 2011). AI systems often provide their scores or "best match" rankings at the beginning of the evaluation process, creating powerful

anchors that shape all subsequent assessments. Wilson et al.'s (2025) findings also reflect this dynamic: participants' decisions clustered around the AI's initial ranking, and they adjusted only slightly—even when the AI's ranking was deliberately biased. Anchoring therefore works in tandem with automation bias. Once the AI establishes the first impression, evaluators both stay close to that anchor and trust it more than their own independent analysis, allowing flawed algorithmic judgments to steer final hiring decisions.

A third mechanism contributing to failures in AI-assisted hiring is overconfidence, which reduces decision makers' willingness to question AI advice. Kahneman (2011) explains that subjective confidence arises from the coherence of the story people construct, not the accuracy of the evidence supporting it. This helps explain why hiring managers may feel certain that they can “spot” algorithmic errors when, in practice, they rarely do. The University of Washington study (Milne, 2024) illustrates this dynamic vividly: large language models favored white-associated names 85% of the time, yet such disparities often went unnoticed because users assumed they were evaluating candidates fairly. This pattern demonstrates overconfidence in two ways. First, evaluators overestimated their ability to detect algorithmic bias, believing they would notice discrimination if it occurred. Second, they overestimated the accuracy and neutrality of the AI system itself, assuming that a data-driven tool would naturally avoid such errors. In both cases, confidence was based on intuition and perceived coherence rather than evidence—precisely the form of overconfidence Kahneman describes. Overconfidence therefore compounds both anchoring and automation bias by creating a false sense of correctness that discourages scrutiny.

Finally, these problems are intensified by the illusion of control, which occurs when people believe they are exercising meaningful oversight even though AI systems have already shaped their decision environment. Rosenthal-von der Pütten and Sach (2024) provide strong

evidence of this mechanism. In their hiring simulation, an AI system systematically favored German-sounding names over Turkish ones by adjusting candidate scores. Participants believed they were independently evaluating resumes and making merit-based decisions, yet approximately 60% failed to detect this systematic bias, even when explicitly asked whether they noticed any favoritism. This reflects a clear illusion of control: evaluators felt they were in charge, but the AI had already structured the information in ways that nudged them toward biased outcomes. Even those who did recognize bias remained constrained by the AI-defined scoring framework. Because the algorithm determined the ordering and emphasis of candidate information, evaluators' decisions were shaped long before they engaged with the resumes—creating the impression of oversight without genuine influence. These design choices reinforce the illusion that humans are overseeing AI systems when, in reality, the system is quietly directing human judgment.

Taken together, these four mechanisms—automation bias, anchoring, overconfidence, and the illusion of control—demonstrate why AI-assisted hiring often fails to achieve fairness or accuracy. They show that technological improvements alone are insufficient; meaningful progress requires recognizing and addressing the psychological processes that shape how humans interact with AI recommendations. Understanding these decision-science mechanisms clarifies why current solutions fall short and points directly to what must change: both hiring systems and the humans who use them need structures that counteract these predictable biases. The next section outlines concrete suggestions designed to translate these insights into more transparent, accountable, and fair hiring practices.

Section IV: Concrete Suggestions

Improving AI assisted hiring requires addressing not only technical flaws in algorithms but also the predictable psychological tendencies that shape how humans interact with these systems. Biases such as automation bias, anchoring, overconfidence, and the illusion of control play a central role in why current uses of AI in hiring processes fall short. The first necessary change targets automation bias. Increasing transparency in how AI systems operate is one of the most effective ways to reduce this tendency. Research by Kupfer et al. (2023) demonstrates that when users are explicitly informed about the possibility of AI errors and are provided with more detailed, less aggregated information, they become less likely to blindly follow automated output and more inclined to engage in deliberate evaluation. Instead of presenting a single overall score or ranking, AI tools should offer audit trails that show how recommendations were generated, including which resume features were emphasized and where uncertainty exists. Making the algorithm's reasoning visible disrupts the perception that AI output is inherently authoritative and prompts evaluators to verify rather than accept machine judgments. By replacing opaque summaries with transparent, interpretable explanations, organizations can meaningfully reduce automation bias and foster more thoughtful, accountable hiring decisions. Requiring brief written justifications when evaluators choose to follow an AI recommendation—especially in rejection decisions—can further slow fast, automatic acceptance and promote more thoughtful review. However, a practical barrier is that its effectiveness still depends on whether recruiters choose to engage with these extra steps rather than bypass them.

A second improvement should require evaluators to conduct blind initial reviews of resumes or interview responses before any AI scores are shown. Research shows that early numerical information creates powerful anchoring effects that shape later judgments even when evaluators know the anchor may be unreliable. In class demonstration on how many BCN majors

are in the class showed that once individuals are exposed to a numerical anchor, their subsequent estimates shift systematically toward that value, and most fail to fully adjust away from it. This illustrates how deeply anchoring shapes human judgment. By removing AI scores from the earliest stage of evaluation, employers would directly weaken this anchoring effect and simultaneously reduce automation bias by ensuring that reviewers form an independent initial assessment before encountering machine outputs. Although recruiters might worry that this step adds time to the workflow, the long-term benefits—more accurate, independent, and defensible hiring decisions—clearly justify the change.

Another change is to reduce overconfidence by implementing structured feedback and calibration training. Organizations should provide feedback showing when reliance on AI recommendations led to poor hiring outcomes and when human overrides resulted in better decisions. Regularly highlighting these cases helps recalibrate evaluators' confidence in both human and algorithmic judgment. Soleimani, Intezari, and Pauleen (2022) emphasize that collaboration and knowledge-sharing between HR managers and AI developers are essential for mitigating cognitive biases, as teams better understand sources of error when feedback loops are built into the evaluation process. This approach directly addresses overconfidence by grounding beliefs about AI accuracy in actual performance data rather than intuition, encouraging more realistic trust and more deliberate review of AI-generated recommendations. Calibration training reinforces this shift by presenting concrete examples of AI failures and successful human judgments, reducing evaluators' tendency to overestimate both their ability to detect bias and the algorithm's overall accuracy.

A final improvement is to educate recruiters and ensure that every AI-assisted decision is accompanied by clear reminders of the system's limitations. Overreliance often emerges because

evaluators assume they can “catch” AI mistakes or guide the system’s outcomes, even without understanding how the algorithm works (Passi & Vorvoreanu, 2022). Research shows that simple reminders that AI systems can make mistakes and that evaluators remain responsible for the final decision significantly reduce automation bias by prompting users to question rather than automatically accept AI recommendations (Ruscheimer & Hondrich, 2024). These reminders realign human decision-makers with their actual role in the process, transforming perceived control into actual, informed oversight. This approach also reduces overconfidence, as many recruiters wrongly believe they can detect algorithmic bias when evidence shows they often cannot. By repeatedly emphasizing uncertainty, fallibility, and human accountability, these prompts disrupt the intuitive confidence that fuels both the illusion of control and undue trust in AI recommendations. In short, transparent education and responsibility reminders help decision makers recalibrate their role in the process, making oversight intentional rather than assumed. Organizations can also reduce the illusion of control by implementing structured disagreement reviews. Whenever AI and a human evaluator reach different conclusions, a third reviewer reassesses the case. This process reveals whether the AI systematically disadvantages certain groups and ensures that human oversight remains substantive rather than symbolic, though companies may view the extra resources required to support it as unnecessary.

Together, these recommendations target the cognitive mechanisms that undermine fairness in AI-assisted hiring by reshaping how humans interact with AI, making evaluators more deliberate, independent, and aware of its limits. By redesigning information flow, increasing transparency, adding purposeful friction, incorporating disagreement checks, and providing ongoing feedback, organizations can reduce bias amplification and move toward a more accurate and equitable hiring process.

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